A CASE-STUDY BASED REPORT

ON

Customer Credit Scoring Using XGBoost and Logistic Regression for Credit Risk Prediction

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MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

This case study examines customer credit-risk prediction using Logistic Regression and XGBoost on the Give Me Some Credit dataset provided as cs-training.csv. The target variable, SeriousDlqin2yrs, indicates whether a customer experienced serious delinquency within two years. The workflow implemented in the project includes data inspection, removal of an unnecessary index column when present, duplicate removal, median imputation for MonthlyIncome and NumberOfDependents, exploratory data analysis, model training, and comparison using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Logistic Regression serves as a baseline model, while XGBoost captures more complex nonlinear patterns. In the current experiment, XGBoost performs better across the reported metrics, although its recall remains limited, showing that many delinquent customers are still missed.

1. Introduction

1.1 Background

Credit scoring supports financial institutions in estimating the likelihood that a borrower will experience serious repayment difficulty. Historical customer information can be analyzed to identify patterns associated with credit risk. This project treats the task as a supervised binary classification problem and compares a linear statistical model with a gradient-boosted tree ensemble.

1.2 Motivation for the Project

A high overall accuracy can be misleading when serious delinquency is relatively uncommon. It is therefore useful to compare Logistic Regression and XGBoost with several metrics rather than accuracy alone. The project also provides practice in preprocessing real-world data, exploring feature distributions, training models, and interpreting classification results.

1.3 Overview of the Work Carried Out

- Loaded cs-training.csv and inspected dimensions, columns, data types, summary statistics, and missing values.
- Removed the Unnamed: 0 column if present and removed duplicate rows.
- Filled missing MonthlyIncome and NumberOfDependents values with their respective medians.
- Explored target distribution, age, monthly income, debt ratio, revolving credit utilization, and feature correlations.
- Split data into training and test sets using an 80:20 ratio and stratified sampling.
- Trained Logistic Regression on standardized features and XGBoost on the unscaled feature matrix.
- Compared models using Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrices, ROC curves, and XGBoost feature importance.

1.4 Scope and Responsible-Use Statement

The project is an educational case study and should not be used as the sole basis for real loan approval, rejection, pricing, or customer treatment. Credit decisions can affect people materially and require appropriate validation, fairness assessment, explanation, privacy protection, and human oversight. The model output is a prediction based on historical data, not a guarantee of future behavior.

2. Problem Statement

2.1 Case-Study Scenario

A financial institution wants to use historical customer records to flag accounts that may experience serious delinquency within the next two years. The project evaluates whether Logistic Regression or XGBoost provides more useful predictions from available customer attributes.

2.2 Central Case-Study Question

How effectively can Logistic Regression and XGBoost predict serious delinquency within two years, and which model provides the better balance of overall discrimination and identification of high-risk customers on the held-out test set?

2.3 Formal Problem Definition

Given a feature vector X containing customer attributes, predict the binary target y = SeriousDlqin2yrs, where 1 indicates serious delinquency within two years and 0 indicates no recorded serious delinquency within that period. Performance is assessed on a held-out test set using classification metrics.

2.4 Objectives that Guide the Study

Understand the dataset structure, target variable, and data-quality issues.

Perform the preprocessing steps implemented in the project.

Use EDA to inspect target imbalance and distributions of selected attributes.

Train Logistic Regression and XGBoost classifiers.

Compare models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Discuss model limitations, especially the ability to identify delinquent customers.

2.5 Assumptions and Limitations

The dataset target label is treated as the ground-truth outcome for this exercise.

The test split is assumed to represent unseen examples from a similar data distribution.

Median imputation is applied only to MonthlyIncome and NumberOfDependents as in the submitted code.

The workflow does not include explicit class rebalancing, threshold optimization, probability calibration, or hyperparameter search.

Historical associations may not remain stable over time; performance may change on new populations or economic conditions.

Feature importance indicates model usage, not causation, and does not by itself establish fairness.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project uses cs-training.csv, associated with the Give Me Some Credit credit-risk prediction task. The target variable is SeriousDlqin2yrs. Predictors include revolving credit utilization, age, past-due-payment counts, debt ratio, monthly income, number of open credit lines and loans, real-estate loans or lines, and number of dependents. Confirm the final dimensions from the dataset output after cleaning.

3.2 Purpose of EDA

Exploratory data analysis was used to understand class distribution, inspect numerical patterns, identify missing data, and review relationships between predictors. The code generates a target count plot, histograms for age, MonthlyIncome, DebtRatio, and RevolvingUtilizationOfUnsecuredLines, and a correlation heatmap.

3.3 Preparation Steps Actually Performed

Read the CSV using pandas.

Inspected shape, columns, data types, descriptive statistics, and missing-value counts.

Dropped Unnamed: 0 if it existed.

Removed duplicate rows.

Imputed missing MonthlyIncome and NumberOfDependents values with medians.

Created X by dropping SeriousDlqin2yrs and used SeriousDlqin2yrs as y.

Used train_test_split with test_size=0.20, random_state=42, and stratify=y.

Applied StandardScaler to training predictors and transformed test predictors for Logistic Regression; XGBoost used the original predictors.

3.4 EDA Exploration – Implementation

The notebook uses pandas and NumPy for tabular analysis, Matplotlib and Seaborn for charts, and scikit-learn for preprocessing and evaluation. Implemented visualizations include:

Count plot of SeriousDlqin2yrs to inspect target-class imbalance.

Histograms of age, MonthlyIncome, DebtRatio, and RevolvingUtilizationOfUnsecuredLines.

Correlation heatmap of numeric columns.

3.5 Summary of EDA Findings

The code is designed to show class balance and distributions of important numeric variables; chart-specific findings should be added after reviewing the rendered plots. The dataset contains missing values in MonthlyIncome and NumberOfDependents, which the code imputes using medians. Because serious delinquency is a minority outcome in this dataset, accuracy alone is not sufficient evidence of model quality.

4. Algorithm Selection and Implementation

4.1 Logistic Regression

Logistic Regression is the baseline classifier. The submitted code configures max_iter=1000 and random_state=42, trains the model on standardized features, and generates class predictions and predicted probabilities. It is comparatively simple but may not capture complex nonlinear patterns.

4.2 XGBoost

XGBoost is a gradient-boosted decision-tree classifier. The submitted configuration uses n_estimators=200, max_depth=4, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, random_state=42, and eval_metric='logloss'. It is trained on the original predictor values.

4.3 Evaluation Metrics

Accuracy: proportion of all test records classified correctly.

Precision: proportion of predicted delinquent cases that are actually delinquent.

Recall: proportion of actual delinquent cases identified by the model.

F1-score: harmonic mean of precision and recall.

ROC-AUC: ability to rank positive cases above negative cases across thresholds.

Confusion matrix: counts of true negatives, false positives, false negatives, and true positives.

4.4 Reported Model Results

The following values were reported for the project; verify them against the final executed notebook before submission.

Metric	Logistic Regression	XGBoost
Accuracy	93.31%	93.74%
Precision	51.30%	59.48%
Recall	3.95%	20.68%
F1-score	7.33%	30.69%
ROC-AUC	69.90%	86.50%

4.5 Discussion

XGBoost performs better than Logistic Regression across the five reported metrics, with a larger ROC-AUC and improved recall. Nevertheless, XGBoost recall of 20.68% means many actual delinquent customers are not identified at the default threshold. Further work could examine class weighting, resampling applied only to training data, decision-threshold tuning based on business costs, probability calibration, cross-validation, and external validation. Changes should be evaluated on data not used to fit or tune the model.

5. Conclusion

This case study implemented a credit-risk classification workflow using Logistic Regression and XGBoost on cs-training.csv. The project included data inspection, duplicate removal, median imputation for two columns, exploratory visualizations, a stratified 80:20 train-test split, model training, and evaluation with five metrics. Based on the reported results, XGBoost achieved higher accuracy, precision, recall, F1-score, and ROC-AUC than Logistic Regression. However, both models—especially at default decision thresholds—have limitations in detecting delinquent customers. XGBoost is the stronger model in this experiment, but further validation, threshold analysis, fairness review, and human oversight are required before real-world use.

6. References

Give Me Some Credit dataset, file used in this project: cs-training.csv. Add the original dataset source/access details used by the team.

Scikit-learn documentation: LogisticRegression, StandardScaler, train_test_split, and classification metrics. https://scikit-learn.org/

XGBoost documentation: XGBClassifier. https://xgboost.readthedocs.io/

Signature of the Team Members

Name: _____ Signature: _____

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